

Basis and Dimension

Abhishek Sarkar

Department of Mathematics
IIT Jodhpur

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Outline

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Linear Dependence

Definition (Linearly Dependent)

Vectors $v_1, \dots, v_k$ are **linearly dependent** if

$$\alpha_1 v_1 + \dots + \alpha_k v_k = 0$$

for some scalars not all zero.

$$\left. \begin{aligned} a+b &= 0 \\ a-b &= 0 \end{aligned} \right\} \Rightarrow a=0=b$$

- **Interpretation:** At least one vector is redundant.

$$\begin{matrix} (2,2) & (1,1) \\ v_1 & v_2 \end{matrix}$$

$$(2,2) = 2(1,1)$$

$$\Rightarrow v_1 = 2v_2 \Rightarrow \boxed{v_1 - 2v_2 = 0}$$

$$\{v_1, \dots, v_k\}$$

$$v_i \rightarrow \alpha_i \neq 0$$

$$\Rightarrow \{v_1, v_2, \dots, v_{i-1}, v_{i+1}, \dots, v_n\}$$



l.i

$$\begin{aligned} a(1,1) + b(1,-1) &= (0,0) \\ \Rightarrow (a,b) + (b,-b) &= (0,0) \\ \Rightarrow (a+b, a-b) &= (0,0) \\ \Rightarrow a+b &= 0 \\ a-b &= 0 \end{aligned}$$

$$\{v_1, v_2, \dots, v_{i-1}, v_i, v_{i+1}, \dots, v_k\} \longrightarrow \text{lin. dep.}$$

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_{i-1} v_{i-1} + \alpha_i v_i + \alpha_{i+1} v_{i+1} + \dots + \alpha_k v_k = \vec{0}$$

Suppose $\alpha_i \neq 0$.

$$\Rightarrow v_i = -\frac{\alpha_1}{\alpha_i} v_1 - \frac{\alpha_2}{\alpha_i} v_2 - \dots - \frac{\alpha_{i-1}}{\alpha_i} v_{i-1} - \frac{\alpha_{i+1}}{\alpha_i} v_{i+1} - \dots - \frac{\alpha_k}{\alpha_i} v_k$$

$$\beta_j = -\frac{\alpha_j}{\alpha_i}$$

$$\Rightarrow v_i = \beta_1 v_1 + \beta_2 v_2 + \dots + \beta_{i-1} v_{i-1} + \beta_{i+1} v_{i+1} + \dots + \beta_k v_k$$

$$\{v_1, \dots, v_{i-1}, v_i, v_{i+1}, \dots, v_k\} \longleftrightarrow \{v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_k\}$$

Linear Independence

Definition (Linear Independence)

Vectors $v_1, \dots, v_k$ are **linearly independent** if

$$\alpha_1 v_1 + \dots + \alpha_k v_k = 0 \Rightarrow \alpha_1 = \dots = \alpha_k = 0.$$

- **Interpretation:** No vector can be written as a combination of others.

Important Facts

$\{v_1, v_2, \dots, v_k, 0\}$
 $(k+1) \rightarrow 0 \cdot v_1 + 0 \cdot v_2 + \dots + 0 \cdot v_k + 1 \cdot 0 = 0$

$0 \begin{pmatrix} 1, 0 \end{pmatrix} + 0 \begin{pmatrix} 0, 1 \end{pmatrix} + 1 \begin{pmatrix} 0, 0 \end{pmatrix} = \begin{pmatrix} 0, 0 \end{pmatrix}$



Theorem

Any set containing 0 is linearly dependent.

Theorem

Given m vectors in $\mathbb{R}^n$, if $m > n$ they are linearly dependent.

$(1, 0), (0, 1), (-1, 1)$

Theorem

A linearly independent set in $\mathbb{R}^n$ has at most n vectors.

$(1, 1), (2, 2) \in \mathbb{R}^2$

Basis of a Vector Space

$$\{(1,0,0), (0,1,0), (0,0,1)\}$$

Definition (Basis of a Vector Space)

A set $\{v_1, \dots, v_n\}$ is a **basis** of V if:

- it spans V
- it is linearly independent

• Every vector in V has a *unique* representation in terms of a basis. (Result)

$x \in V$.

$$x = \alpha_1 v_1 + \dots + \alpha_k v_k$$

Suppose it has another representation.

$$x = \beta_1 v_1 + \dots + \beta_k v_k$$

$$\alpha_1 v_1 + \dots + \alpha_k v_k = \beta_1 v_1 + \dots + \beta_k v_k$$

$$\Rightarrow (\alpha_1 - \beta_1) v_1 + \dots + (\alpha_k - \beta_k) v_k = 0$$

$$\text{e.i.} \Rightarrow \alpha_1 - \beta_1 = 0 = \alpha_2 - \beta_2 = \dots = \alpha_k - \beta_k$$

$W \supset S$.
 $\Rightarrow W$ is a subspace in V .
 $W \supset \text{span}(S)$



$\text{span}(S)$ is the smallest subspace containing S .

$\{(1,0), (0,1)\}$ will span $\mathbb{R}^2$

$\{(1,0), (0,1), (1,-1)\}$ will also span $\mathbb{R}^2$.

W spans $\mathbb{R}^2$

$W \subseteq X \subseteq \mathbb{R}^2$.

W spans V .

and $W \subseteq X \subseteq V$.

$\Rightarrow X$ will also span V .

Examples of Bases

$$\{ \underbrace{(1, 0, 0, \dots, 0)}_{e_1}, \underbrace{(0, 1, 0, \dots, 0)}_{e_2}, \dots, \underbrace{(0, 0, 0, \dots, 1, 0, \dots, 0)}_{e_i}, \dots, \underbrace{(0, 0, \dots, 0, 1)}_{e_n} \}$$

- Standard basis of $\mathbb{R}^n$
- $\{1, x, x^2, \dots, x^n\}$ for polynomials of degree $\leq n$



$$\{ (2, 0, 0, \dots, 0), (0, -1, 0, \dots, 0), \dots, (0, 0, \dots, -\frac{1}{2}, 0, \dots, 0) \}$$

$$(a, b) = a(1, 0) + b(0, 1)$$

Dimension

Definition (Dimension)

The **dimension** of V , denoted $\dim V$, is the number of vectors in a basis of V .

Theorem

Any two bases of a finite-dimensional vector space have the same number of elements.

Examples

basis for $P_n \rightarrow \{1, x^1, x^2, \dots, x^n\}$
↑
n.

- $\dim(\mathbb{R}^n) = n$
- $\dim(\mathcal{P}_n) = n + 1$ ←
- $\dim(\{0\}) = 0$
←

Subspaces and Dimension

Theorem

If W is a subspace of a finite-dimensional vector space V , then

$$\dim W \leq \dim V.$$

Equality holds if and only if $W = V$.

Basis and coordinates

- If $\{v_1, v_2, \dots, v_n\}$ is a basis for a vector space V , then any vector $v \in V$ has a unique representation

$$v = x_1 v_1 + x_2 v_2 + \dots + x_n v_n,$$

where $x_i \in \mathbb{R}$.

- The coefficients $x_1, x_2, \dots, x_n$ are called the coordinates of v with respect to the ordered basis $v_1, v_2, \dots, v_n$.
- The mapping vector $v \mapsto (x_1, x_2, \dots, x_n)$ is a one-to-one correspondence between V and $\mathbb{R}^n$. This correspondence respects linear operations in V and in $\mathbb{R}^n$.

Some Important Results on Span

Remark

In general

1. Any set of vectors in $\mathbb{R}^2$ which contains two non colinear vectors will span $\mathbb{R}^2$.
2. Any set of vectors in $\mathbb{R}^3$ which contains three non coplanar vectors will span $\mathbb{R}^3$.
3. Two non-colinear vectors in $\mathbb{R}^3$ will span a plane in $\mathbb{R}^3$.

Example

The subspace of $\mathbb{R}^3$ spanned by $v_1 = (1, 0, 1)$ and $v_2 = (0, 1, 1)$ is the plane with Cartesian equation $z = x + y$.

Several Examples

Example (Basis and Coordinates ◦ Euclidean Space)

Coordinates of a vector $v = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ relative to the standard basis $e_1 = (1, 0, \dots, 0, 0)$, $e_2 = (0, 1, \dots, 0, 0)$, $\dots$, $e_n = (0, 0, \dots, 0, 1)$ are $(x_1, x_2, \dots, x_n)$.

Example (Basis and Coordinates ◦ Matrix Spaces)

Coordinates of a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathcal{M}_{2,2}(\mathbb{R})$ relative to the basis $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ are (a, c, b, d) .

Example (Basis and Coordinates ◦ Polynomial Space)

Coordinates of a polynomial $p(x) = a_0 + a_1x + \dots + a_{n-1}x_{n-1} \in \mathcal{P}_n$ relative to the basis $1, x, x^2, \dots, x_{n-1}$ are $(a_0, a_1, \dots, a_{n-1})$

Examples of Coordinates With Respect to a Basis

- Vectors $u_1 = (2, 1)$ and $u_2 = (3, 1)$ form a basis for $\mathbb{R}^2$.

Problem

Find the coordinates of the vector $v = (7, 4)$ with respect to the basis u_1, u_2 .

Thank You

Questions?